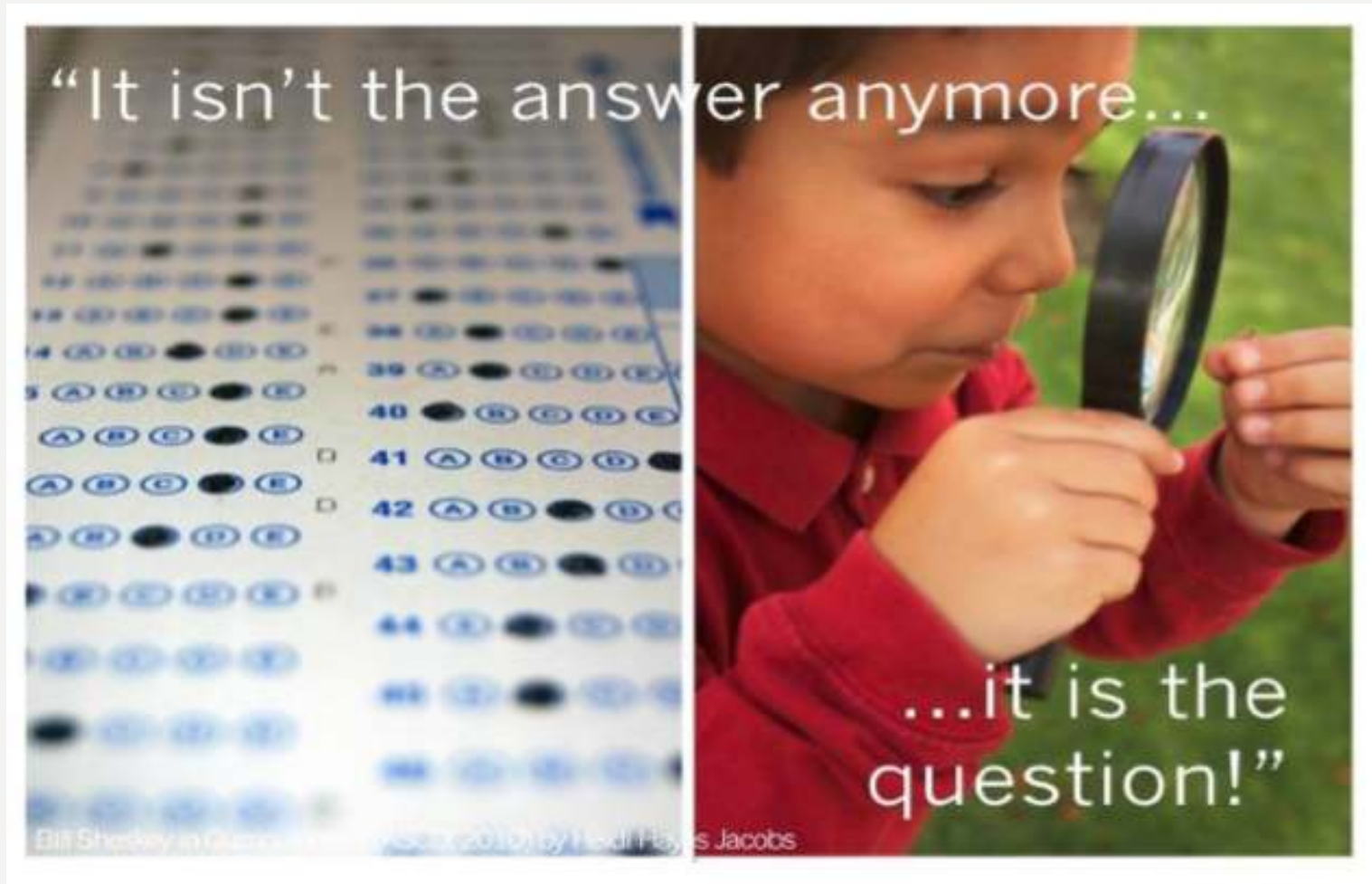




KEY INQUIRY QUESTIONS

APRIL, 2019

FROM ANSWER TO QUESTION.....





FOUNDATION FOR KEY INQUIRY QUESTION

The foundation for key inquiry question (KIQ) is Inquiry based learning.

- Inquiry Based learning involves experiences that enable learners to develop understanding about aspects of the world around them through the development and use of investigative skills.

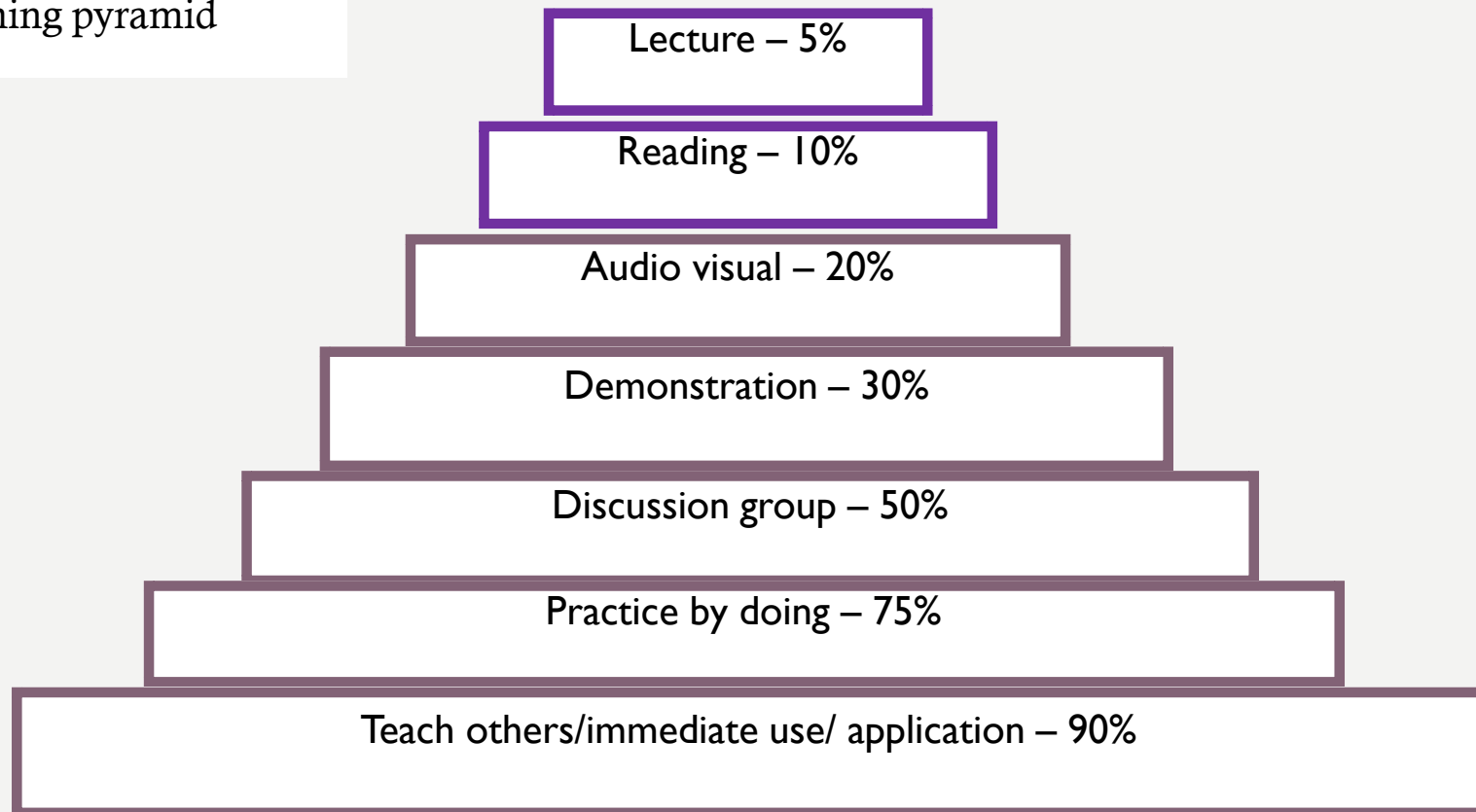
incorporates a process of exploring the world, which leads to:

- a) Asking questions
 - b) Making discoveries
 - c) Conducting research/ experiments
- Its philosophy is based on constructivist learning theories (Dewey, Piaget, Vygotsky) whereby humans construct knowledge and meaning from their experiences.



WHY IBL?

learning pyramid





IMPORTANCE OF IBL?

- Learning is about preparing learners for a world that is **dynamic not fixed and static**.
- Education cannot give learners all the information that they need to know, but rather it must provide the **tools for continuation in learning**.
- Inquiry learning and active learner involvement lead to **realization of learning outcomes** in the classroom/ learning environment.
- Learners who actively make observations, collect, analyse ,synthesise information and draw conclusions develop **problem-solving skills** that can be applied to future '**need to know**' situations.



INQUIRY BASED INSTRUCTION

- More **learner centred**, with the **teacher as a facilitator** of learning.
- Emphasis on '**how we come to know**' and less on '**what we know**'.
- Concerned with in-school success equally with preparation for **life-long learning**.
- Open systems where learners are encouraged to search and make use of resources **beyond the classroom and the school**.
- Assessment is focused on determining the **progress of skills development** in addition to understanding content.
- **Uses technology** to connect learners appropriately with local and world communities which are rich sources.
- Focus on the **how** instead of the **what**.

So....What kind of questions do we ask to facilitate learning???



WHAT IS A QUESTION?

Activity

From your learning area, eg. Math or English can you write some of the questions you have asked learners in class? May be done in groups of 2 or 3.

Brainstorm on the nature of the questions into: closed vs open-ended questions.

Brainstorm on inquiry type of questions.



WHAT KIND OF QUESTIONS???

- These are questions that help to focus learning.
- Questions that probe for deeper meaning and set the stage for further questioning
- Foster the development of critical thinking skills and higher order capabilities such as problem solving.



WHAT MAKES A GOOD KEY INQUIRY QUESTION

- It is open-ended, non-judgmental, meaningful and purposeful with an aim to allow learners to explore ideas.
- Is thought-provoking and intellectually engaging, often sparking discussion and debate.
- Encourages collaboration amongst learners, teachers, and the community
- Integrates technology to support the learning process.



CONTD...

- Call for higher-order thinking, such as analysis, inference, evaluation, prediction. It cannot be effectively answered by recall alone.
- Point toward important, transferable ideas within and even across subjects.
- Raise additional questions and sparks further inquiry.
- Requires support and justification, not just an answer.



WHAT A KEY INQUIRY QUESTION IS NOT

1. It is **NOT** a direct question that can be answered with a yes or no.
2. It is **NOT** a question that can be answered without support.
3. It is **NOT** blurry/vague (undefined) questions.

Activity:

Ask participants to discuss in pairs and give examples of questions in each area



FORMING AN INQUIRY QUESTION

- Consider the strands and sub-strands, and specific learning outcomes as given in the curriculum design.
- examine the theme or concept in the curriculum that needs to be addressed.

Activity:

choose a learning area and brainstorm on questions that would enable learners to think about the concept without dictating the direction or outcome of their thinking



CONTD...

Utilize the six typical question words:

- Why? How?
- Who?
- Where?
- When?
- What?



RELATED QUESTIONS....

Once an inquiry question has been formed, the next step is to formulate a list of related questions that assist the learner in answering the key question.



EXAMPLE 1

Grade 1: Mathematics activities

Strand: Numbers, sub-strand: number concept.

- Specific Learning Outcome:

By the end of the sub-strand, the learner should be able to:

a) sort and group objects according to different attributes within the classroom,

*Nb: use simpler terms that learners know and understand. Eg. **Sort and group**, use **arrange**. **Object** use the word **things**. For the word **attributes** use 'the way things are', what is same or different.*

Question

Discuss the questions that we would ask learners before we start the lesson on the given sub-strand, learning outcome.



NAME SOME OF THE OBJECTS TO SORT AND GROUP IN CLASSROOM

Common objects in the classrooms

- ✓ writing materials
- ✓ furniture
- ✓ **Waste materials**
- ✓ Cleaning materials



INQUIRY QUESTION
HOW CAN WE SORT OUT THE WASTE IN THE CLASSROOM?

Learners to discuss in groups of 4 and organize how they will be sorting out waste in the classroom.

Learners to carry out a classroom based project by having buckets labelled for different waste materials according to the attribute they have chosen.

Key words: sort, group, objects, attributes



ATTRIBUTES FOR SORTING OUT WASTE

- Recycling/ re-use
- Burning
- Decompose/ rotting



EXAMPLE 2: LEARNING OUTCOME

Appreciate the use of sorting and grouping items in day to day activities.

Activity

In pairs write the questions that you will ask learners as you carry out the lesson on the given sub-strand, learning outcome.



KEY INQUIRY QUESTION

- How do you sort and group (arrange) different things at home?
(this activity to be carried out as a project with the help of the parents/ care givers).

Parents to guide the learner in giving explanations on different ways of sorting and grouping things at home. (parental engagement and empowerment)

- *Other places where the concept is applicable: market place, grocery stores, malls etc. (CSL, non-formal activity).*

Activity

Participants to work in groups, brainstorm and generate possible key inquiry questions in different learning areas and grades.



EXAMPLE 3: GRADE 4

LEARNING AREA: MATHEMATICS

Strand : Measurement

Sub-strand- length

- Learning outcome:

(f) work out perimeter of plane figures in different context

What is a plane figure? Participants to brainstorm and give responses.

In groups of 3 or 4 identify different plane figures in the room.

*How do we work out the **perimeter** of a given plane figures?*

Activity

Participants to work out perimeter of a given plane figure in real life. Eg. A room floor, piece of paper, circular plane eg. Plate, etc.

Key feature: addition of the length of all sides.



TYPES OF PLANE FIGURES

Examples of real things that have plane figure shapes

Rectangles, floors, tiles, ceiling, pieces of papers,

Squares,

Circles, pools, tiles, patterns in clothes, curtains, plates,

Triangles , patterns in garments etc

Others????



SNE: LEARNING AREA ACTIVITIES OF DAILY LIVING SKILLS

- Strand: personal items
- Sub-strand: Grooming items
- Key Inquiry question:

Ordinary direct questions

Can you name some of the grooming items we use?

What grooming items do we use in the morning?



INQUIRY QUESTIONS

1. How do you make yourself smart?
2. what do you do to look smart?



AN EXAMPLE IN HISTORY/SOCIAL STUDIES: DOCUMENT BASED RESEARCH ON THE KAPENGURIA SIX





QUESTION

**WHO AMONG THE PEOPLE IN THE PHOTO WAS NOT ONE OF THE
'KAPENGURIA SIX'?**

Learners research from sources provided by the teacher/ facilitators:

Books,

Newspapers,

Interview of key informants,

videos, etc



EXAMPLES OF KEY INQUIRY QUESTIONS

Art and Dance

- In what ways does art reflect culture as well as shape it?
- Where do artists get their ideas?
- How do we use dance to communicate?



LANGUAGES

- What makes a great story/book?
- Should a story teach its reader something?
- Why do we punctuate? What might happen if we didn't have punctuation marks?
- Why do you write?
- What do good readers do when they don't understand what they are reading?



SCIENCE

- How can we take care of the environment?
- Why should we take care of the environment?
- How do human beings and animals survive in harsh or changing environments?
- Why is washing hands important?



OTHERS SUBJECTS

- **Physical Education**
- Who is a winner?
- Is pain necessary for progress in athletics? (“No pain, no gain.”)
- **Technology**
- In what ways can technology enhance expression and communication?
- In what ways might technology hinder it?



CHARACTERISTICS OF TYPES OF CLASSROOM QUESTIONS

- **Questions That Hook**
- **Questions That Lead**
- **Questions That Guide**
- **Questions that are Essential (KIQ)**



QUESTIONS THAT HOOK

- Asked to interest learners around a new topic
- May spark curiosity, questions, or debate
- Asked once or twice, but not revisited



QUESTIONS THAT LEAD

- Asked to be answered
- Have a "correct" answer
- Support recall and information finding
- Asked once (or until *the* answer is given)
- Require no (or minimal) support



QUESTIONS THAT GUIDE

- Asked to encourage and guide exploration of a topic
- Point toward desired knowledge and skill (but not necessarily to a single answer)
- May be asked over time
- Generally require some explanation and support



KEY INQUIRY QUESTIONS

- Asked to stimulate ongoing thinking and inquiry
- Raise more questions
- Spark discussion and debate
- Demand justification and support
- Responses may change as understanding deepens



CONCLUSION

Michelle Obama in her book ' Becoming, 2018' says

*“My mother..... **Showed** me **how to think** for myself and to **use** my voice”*

John Dewey

*“Give the pupils something **to do**, not something to learn; and the **doing is of such a nature as to demand thinking**; learning naturally results.”*



END

THANK YOU